CHOW'S LEMMA FOR SEPARATED FINITE-TYPE MORPHISMS WITH FINITELY MANY COMPONENTS (EGA ROUTE)

THE STACKS PROJECT AUTHORS

Editorial note. All steps explicit; the alternative 'standard proof' route is mentioned in the source but not required.

Allowed standard prerequisites (editorial).

- Noetherian Chow lemma. Cohomology of Schemes, Lemma coherent-lemma-chow-Noetherian

Source reference. Original reference 0200: sources/stacks-chow/tags.txt, source lines 2617–2617. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01ZX: sources/stacks-chow/tags.txt, source lines 2612–2612. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0202: sources/stacks-chow/tags.txt, source lines 2619–2619. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 01ZJ: sources/stacks-chow/tags.txt, source lines 2596–2596. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZQ: sources/stacks-chow/tags.txt, source lines 2603–2603. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01YZ: sources/stacks-chow/tags.txt, source lines 2574–2574. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01YY: sources/stacks-chow/tags.txt, source lines 2573–2573. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01R7: sources/stacks-chow/tags.txt, source lines 2292–2292. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 01R8: sources/stacks-chow/tags.txt, source lines 2293–2293. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01RL: sources/stacks-chow/tags.txt, source lines 2306–2306. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $f : X \to S$ be a separated morphism of finite type. Assume that X has finitely many irreducible components. Then there exists an $n \geq 0$ and a diagram*

$$\begin{array}{ccccc} X & \xleftarrow{\pi} & X' & \longrightarrow & \mathbf{P}^n_S \\ & \searrow & \downarrow & \swarrow & \\ & & S & & \end{array}$$

where $X' \to \mathbf{P}^n_S$ is an immersion, and $\pi : X' \to X$ is proper and surjective. Moreover, there exists an open dense subscheme $U \subset X$ such that $\pi^{-1}(U) \to U$ is an isomorphism of schemes.

Proof. Let $X = Z_1 \cup \ldots \cup Z_n$ be the decomposition of X into irreducible components. Let $\eta_j \in Z_j$ be the generic point.

There are (at least) two ways to proceed with the proof. The first is to redo the proof of Cohomology of Schemes, Lemma 0200 using the general Properties, Lemma 01ZX to find suitable affine opens in X. (This is the "standard" proof.) The second is to use absolute Noetherian approximation as in the proof of Lemma 0202 above. This is what we will do here.

By Proposition 01ZJ we can find a closed immersion $X \to Y$ where Y is separated and of finite presentation over S. Write $S = \lim_i S_i$ as a directed limit of Noetherian schemes, see Proposition 01ZA. By Lemma 01ZM we can find an index $i \in I$ and a scheme $Y_i \to S_i$ of finite presentation so that $Y = S \times_{S_i} Y_i$. By Lemma 01ZQ we may assume that $Y_i \to S_i$ is separated. We have the following diagram

$$\begin{array}{ccccccc} \eta_j \in Z_j & \longrightarrow & X & \longrightarrow & Y & \longrightarrow & Y_i \\ & & & \searrow & \downarrow & & \downarrow \\ & & & & S & \longrightarrow & S_i \end{array}$$

Denote $h : X \to Y_i$ the composition.

For $i' \geq i$ write $Y_{i'} = S_{i'} \times_{S_i} Y_i$. Then $Y = \lim_{i' \geq i} Y_{i'}$, see Lemma 01YZ. Choose $j, j' \in \{1, \ldots, n\}$, $j \neq j'$. Note that η_j is not a specialization of $\eta_{j'}$. By Lemma 01YY we can replace i by a bigger index and assume that $h(\eta_j)$ is not a specialization of $h(\eta_{j'})$ for all pairs (j, j') as above. For such an index, let $Y' \subset Y_i$ be the scheme theoretic image of $h : X \to Y_i$, see Morphisms, Definition 01R7. The morphism h is quasi-compact as the composition of the quasi-compact morphisms $X \to Y$ and $Y \to Y_i$ (which is affine). Hence by Morphisms, Lemma 01R8 the morphism $X \to Y'$ is dominant. Thus the generic points of Y' are all contained in the set $\{h(\eta_1), \ldots, h(\eta_n)\}$, see Morphisms, Lemma 01RL. Since none of the $h(\eta_j)$ is the specialization of another we see that the points $h(\eta_1), \ldots, h(\eta_n)$ are pairwise distinct and are each a generic point of Y'.

We apply Cohomology of Schemes, Lemma 0200 above to the morphism $Y' \to S_i$. This gives a diagram

$$\begin{array}{ccccc} Y' & \xleftarrow{\pi} & Y^* & \longrightarrow & \mathbf{P}^n_{S_i} \\ & \searrow & \downarrow & \swarrow & \\ & & S_i & & \end{array}$$

such that π is proper and surjective and an isomorphism over a dense open subscheme $V \subset Y'$. By our choice of i above we know that $h(\eta_1), \ldots, h(\eta_n) \in V$. Consider the commutative diagram

$$\begin{array}{ccccccc} X' & = & X \times_{Y'} Y^* & \longrightarrow & Y^* & \longrightarrow & \mathbf{P}^n_{S_i} \\ & & \downarrow & & \downarrow & \nearrow & \\ & & X & \longrightarrow & Y' & & \\ & & \downarrow & & \downarrow & & \\ & & S & \longrightarrow & S_i & & \end{array}$$

Note that $X' \to X$ is an isomorphism over the open subscheme $U = h^{-1}(V)$ which contains each of the η_j and hence is dense in X. We conclude $X \leftarrow X' \to \mathbf{P}^n_S$ is a solution to the problem posed in the lemma. $\square$